

Breast Cancer Detection Using Deep Learning and Grad-CAM: A Comparative Analysis of CNN and ResNet Models

By

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Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

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Table of Contents

Acknowledgement	5
Abstract.....	6
Chapter 1: Introduction	7
1.1 Problem Discrimination, Background and Motivation	7
1.2 Objectives.....	8
1.3 Methodology.....	9
1.4 Legal, Social, Ethical and Professional standards	9
1.5 Background	10
1.6 Structure of Report	11
Chapter 2: Literature Review	11
2.1 Literature Review.....	11
2.2 Technology Review	13
2.3 Summary	14
Chapter 3: Implementation	14
3.1 System Overview / Pipeline	14
3.2 Dataset Description.....	15
3.3 Exploratory Data Analysis (EDA)	16
Table 1: Dataset Distribution	16
Figure 3.1: Distribution of breast ultrasound images per class	16
Figure 3.2: Distribution of the classes in percentage.....	17
Figure 3.4: Image Width and Height Distribution	19
Figure 3.5: Boxplot of the image dimensions	19
Figure 3.6: Image Width by Height Scatter Plot.....	20
Figure 3.7: Image Size Density Distribution	20
Figure 3.8: Net Width vs Height Correlation.....	21
Figure 3.9: Violin Plot of Image Dimensions	21
Figure 3.10: Image Brightness Distribution.....	22
Figure 3.11 Pair Plot of Image Dimensions	22
Figure 3.12: 3D Scatter Plot of Image Dimensions.....	23
Figure 3.13: 3D Density Surface of Sizes of Images	23

Summary of EDA Findings	23
3.4 Data Preprocessing	24
3.5 CNN Model Development	25
3.6 ResNet Model Development.....	27
3.7 Model Training Strategy.....	28
Chapter 4: Evaluation and Results	30
4.1 Performance Metrics Overview	30
4.2 CNN Model Evaluation	30
4.3 ResNet Model Evaluation	32
4.4 Grad-CAM Analysis (Interpretability).....	33
4.5 Comparative Analysis.....	35
Table 2: Model Performance Comparison	35
4.6 Summary of Results	35
Chapter 5: Conclusion	36
5.1 Future Work	37
5.2 Reflection	38
6. References	39
7. Appendices.....	42
1. Github Repository Link:.....	42
2. Dataset Link:.....	42

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Abstract

Breast cancer is the second leading cause of death among women globally, thus timely and accurate detection is crucial for effective treatment and improved survival rates. Medical imaging modalities like ultrasound are critical tools in diagnostic settings, yet their manual interpretation may be tedious and prone to variability. The proposed deep learning-based automated breast cancer detection method using ultrasound images, offers an innovative solution for localization and detection with special focus on improving classification performance as well as model interpretability.

This study used a dataset comprising breast ultrasound images, which were classified into three types: normal, benign and malignant. To facilitate generalizability and between our clinical goals, the dataset is reframed from a multiple-class prediction task into a binary classification task through merging benign and malignant images into one class hence considered as “cancer,” with normal treated as “non-cancer.” The preprocessing pipeline was applied to the dataset, including image resizing, normalization and split into train-test datasets.

In this study, we created and tested two deep learning models: a learn model based on custom-native Convolutional Neural Network (CNN), and the Residual Network (ResNet) based on transfer learning. The CNN model structure was able to extract hierarchical features directly from the dataset, and the ResNet also made use of pre-trained weights to improve feature learning. We trained and evaluated both models on standard performance metrics (accuracy, precision, recall, F1-score confusion matrix and ROC-AUC).

Experimental results show that CNN model outperforms ResNet with 94% accuracy compared to ResNet of age 91%. It also showed higher recall for cancer detection with CNN giving it an edge over the other for medical diagnosis. The ROC-AUC scores that invariably support the high discriminative power of both models, finding their AUC equal to 0.974 in case of CNN.

To improve interpretability, we used Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the regions of interest that drive model decisions. The CNN model generated localized, clinically relevant heatmaps that focused on tumor regions, while the ResNet model tended to produce more diffuse activation maps.

In conclusion, this study indicates that a well-designed CNN model could describe an accurate, reliable and interpretable breast cancer detection. The findings underscore the importance of model selection and explainability in medical AI systems and our experiments show that simpler architectures can be highly effective for datasets specific to a given domain.

Chapter 1: Introduction

1.1 Problem Discrimination, Background and Motivation

Breast cancer is one of the most common and lethal diseases that women suffer throughout the world. Early detection plays a key role in improving survival rates and permitting effective treatment. Breast imaging techniques like ultrasound, mammogram and MRI (magnetic resonance imaging) are widely used in the diagnosis of breast abnormalities. Among these, the ultrasound imaging became widely used because of its cost-effectiveness and safety in detecting tumors in dense breast tissues. Nevertheless, the accurate interpretations of ultrasound images are heavily reliant on radiologist skills which render the processes at this stage of diagnosis subjective and prone to variability.

In the past few years, artificial intelligence (AI) and deep learning in particular has shown great potential in enhancing the accuracy and efficiency of medical image analysis. Convolutional Neural Networks (CNNs) have been used to achieve high performance for image classification tasks, such as breast cancer classification [3], [8]. These models can automatically learn hierarchical representations from images which lessens the requirement for manual feature extraction. Moreover, transfer learning approaches based on pre-trained architectures have also been widely proposed and used in this task to improve performance when working with small medical datasets [1], [5].

Researchers are now dealing with the trade-off of using these powerful models but at a cost, as deep learning black boxes lose interpretability. Deep neural networks are sometimes referred to as “black-box” systems [4], meaning the rationale behind their decisions remains opaque to clinicians. The absence of explainability; in particular, this hinders their use in clinical practice due to the need to build trust and accountability. To mitigate the effect of this drawback, explainable artificial intelligence (XAI) techniques have been proposed to extract interpretability from model predictions. Gradient-weighted Class Activation Mapping (Grad-CAM) is one of the most commonly used approaches. Grad-CAM provides regions or heatmaps which indicate how much a region in an input image contributed to classifying it into the target class [17].

The second consideration is the trade-off between model complexity and interpretability. Although more complex architectures like ResNet can achieve high accuracy as they are capable of expressing level nuances [12], they are not always able to offer interpretable visual explanations. In contrast, simpler CNN architectures may be competitive while providing greater interpretability. This

opens an important research question about the trade-off between predictive performance and explainability in medical image analysis.

Driven by these challenges, this study proposed, developed and evaluates deep learning models for the detection of breast cancer using ultrasound images with an emphasis on interpretability. This paper attempts to bridge the gap further by comparing a custom CNN model against an architecture based on transfer learning, with the help of visual explanation techniques implemented through Grad-CAM, providing critical insights into how varying architectures affect results in terms of performance and interpretability. This work is expected to promote the development of more trusted and interpretable AI models for medical diagnostics.

1.2 Objectives

This project aims at developing a deep learning framework that is accurate and interpretable in detecting breast cancer using ultrasound imaging. In order to meet this objective, several specific aims have been identified.

Primarily, the project focuses on pre-processing and preparing a publicly available breast ultrasound dataset for efficient model training. This entails data preparation, addressing the class labels, and transforming the classification problem into a binary task with benign and malignant cases clustered as cancer while normal cases are classified as non-cancer.

In addition, the project should also create and deploy a CNN model suitable for image classification. This model will be used as the baseline, one that when trained achieves good classification performance with a faithful representation of the data.

the third one will be transfer learning using pre-trained model such as ResNet. The goal is to determine if deeper architectures improve upon classification accuracy compared to the custom CNN model.

Another main goal is to use explainability mechanisms, specifically Grad-CAMs, to visualize which areas of the input ultrasound images contribute more to its prediction. This will allow a comparison of how various models interpret the same input data.

At last, The project through metrics like accuracy, confusion matrix and ROC-AUC will assess the performance of the models together with checking their interpretability. In general the goal is to find a model that achieves a trade-off between strong performance and an interpretable explanation.

1.3 Methodology

The methodology for breast cancer detection is structured in this project. Data preparation and the model development steps are followed by a loop training -> evaluation -> interpretation.

First, a dataset of breast ultrasound images is obtained and processed. This dataset can be classified into three different classes that are 'normal' and 'benign' and 'malignant'. To simplify and improve classification performance, the task is reformulated to a binary classification problem, whereby normal images are classified as non-cancer class and benign/malignant image data is combined into one cancer class. The pictures are resized and normalized in order for them to be consistent with what the deep learning models will accept.

After data preparation, two distinct models are implemented. Model 1: Custom CNN with a number of convolutional and pooling layers to obtain features from images. In the second model, transfer learning is applied on the pre-trained ResNet architecture by using knowledge gained from previous tasks to enhance performance in related problems.

Using relevant optimization techniques like the Adam optimizer, both models are trained and tested on a test dataset. Classification effectiveness is measured using performance metrics such as accuracy, loss, confusion matrix and ROC-AUC.

Grad-CAM is then applied to both models for better interpretability. This technique creates heatmaps which indicate the parts of the image that are most related to prediction. Comparing these visualizations allows the study to assess how much of each model's attention is directed towards clinically significant areas.

1.4 Legal, Social, Ethical and Professional standards

There are many legal, social, ethical or professional issues that must be critically considered for the use of artificial intelligence in health care. When discussing breast cancer detection the patient data privacy and confidentiality is a critical factor. While the dataset used for this project is publicly available, deidentified and anonymized, care must be taken to handle data responsibly and not expose any personally identifiable information.

Data journalists and AI experts worry that the algorithm it uses to make decisions can determine malign intent beyond human reproach. Most deep learning models are black-box, meaning they make predictions but do not explain the logic behind these predictions; since clinical decision-making relies heavily on explanations of what lies behind a prediction, this can be problematic. By

using explainability techniques like Grad-CAM we were able to remedy this and through the ability of visual evidence help word for words a model that is more trusted by healthcare personnel.

For example, if the training dataset does not adequately represent different patient populations, this could result in biased predictions by the model that contribute to healthcare disparities. As a result, it is critically important to recognize dataset constraints and demonstrate that the model undergoes thorough evaluation.

On the professional level, AI systems should be considered decision-support devices instead of substitutes for medical specialists. The conclusive diagnosis must always be made by qualified medical professionals — AI should work as assistive technology to streamline the process and increase accuracy.

1.5 Background

Deep learning has dramatically changed the field of computer vision, especially for image classification and medical image analysis tasks. One of the most popular deep learning models for such application is Convolutional Neural Networks (CNNs). These architectures usually contain a stack of multiple convolutional pairs, pooling layers, and fully connected layers that use core operations to detect hierarchical features [13].

CNNs have been extensively used in medical imaging for various tasks including tumor detection, segmentation, and classification. Because of their capacity to automatically discover relevant features, these methods are particularly well-positioned to aid in analyzing complex patterns in ultrasound images [3]. However, the performance of CNNs in many cases relies heavily on abundant datasets, which can be scarce in medical fields.

Transfer learning has been proposed as a key strategy to tackle this issue. Transfer learning involves using pre-trained models, such as ResNet, which are trained on large scale datasets including ImageNet and fine-tuning them for certain medical tasks [12]. We can train very deep networks easily with negligible degradation problem through the use of residual connections in an architecture called ResNet which has been subsequently shown to be very effective at image classification.

While deep learning models have been spectacularly successful, they are often critiqued for not being interpretable. As a result, Explainable AI (XAI) techniques have been developed to demystify model predictions. One such technique is Grad-CAM, which provides visual explanations by highlighting relevant regions in the input image conditioned on gradient information [17].

Knowledge of these concepts is crucial for building high-performing and interpretable breast cancer detection systems. This project leverages these foundational techniques to consider the relationship between model performance and interpretability.

1.6 Structure of Report

The report is structured around five chapters. Chapter 1 outlines the problem, aims, discussing, and significance of the project. In Chapter 2, a literature and technology review is presented where existing research and tools for detecting breast cancer using deep learning are discussed. Chapter 3 The Implementation Process (dataset preparation, model development and training procedures) In Chapter 4 we describe the evaluation and results (such as performance metrics and Grad-CAM visualizations), together with a comparison between the models. Lastly, Chapter 5 wraps up the research with a conclusion for the findings as well as considerations for future work and some reflections on the project overall.

Chapter 2: Literature Review

2.1 Literature Review

Detection of breast cancer is one of the heavily researched topics in medical imaging because it plays an important role in early diagnosis, and subsequently patient survival. Conventional diagnostic approaches depend on the expertise of radiologists, which can introduce variation in interpretation. To overcome this limitation, in recent years, more and more studies have looked into the use of deep learning techniques specifically Convolutional Neural Networks (CNNs) for automated breast cancer detection.

Early studies showed that CNNs could automatically extract hierarchical features from medical images without the need for hand-crafted feature engineering. Bardou et al. [3] was to classify the histology images of breast cancer, which also reflects a promising classification rate using convolutional neural networks. Similarly, Wang et al. [8] has focused on the use of deep learning with a new approach of identifying metastatic breast cancer and thus demonstrates the powerful ability of CNNs to discover complex patterns in medical data. These studies demonstrated the use of CNNs as a trusted approach to image based diagnosis.

Due to the success of deep learning, transfer learning has gained popularity as a common approach – particularly in cases with small medical data. Saber et al. Axtell et al.[1] developed a model to classify breast cancer based on transfer learning, showing that pre-trained networks significantly

enhance performance. Tang et al. Despite that, [5] conducted additional research examining transfer learning in ultrasound imaging and demonstrated that pre-trained models could adapt well to domain-specific tasks. In addition, Rao et al. Hybridization for improved classification accuracy is used in multi-model combinations with the ensemble transfer learning strategy proposed by [7].

ResNet-like deep architectures have become popular due to their ability to train extremely deep nets without degradation problems [12]. Additionally, the use of residual connections in ResNet helps the model to learn more complex features better, which can also result in improved classification accuracy. Shankar et al. To improve breast cancer detection further, [4] combined deep transfer learning with optimization techniques. Similarly, Kim et al. [9] proposed attention-based CNN models to enhance feature representation in ultrasound images. These methods achieve high accuracy at the cost of increased model complexity.

Despite these improvements, one of the major pitfalls that deep learning models in medical imaging still face is the non-interpretable nature of their solutions. While the majority of studies focus on maximizing classification accuracy, little attention is paid to their interpretability. The absence of transparency presents a major hurdle for clinical applications, where explainability is critical for building trust and ensuring validation.

To overcome this problem, Explainable Artificial Intelligence (XAI) techniques are proposed. Selvaraju et al. [17] proposed a gradient-based visualization method called Grad-CAM, which shows the most important regions in an image that help make a model's prediction. Grad-CAM has been commonly used in medical imaging tasks for visualisation outputs as well as transparency of the underlying model. The conversation around interpretability in deep learning was continued by Zeiler and Fergus [16], who also presented a means of visualization to understand convolutional networks.

Moreover, survey studies like Litjens et al. [14] and Roslidar et al. It should be noted that [1], [2] provide extensive surveys on deep learning for medical imaging, mentioning strengths and weaknesses of existing methods. These studies highlight that despite the accuracy potential of deep learning models, lack of interpretability is a major roadblock for clinical adoption.

Some more recent trend in the literature is to use EfficientNet, a next generation architecture that leverages efficient model scaling leading with better performance [10]. While state-of-the-art results can be achieved using such models, they also introduce additional complexity in deep learning systems and thus make deep learning systems less interpretable.

Such literature shows that most of the work is focused on getting high accuracy using deep architectures and optimized models. Yet there is a stark lack of assessment in the trade-off between

model performance and interpretability. Specifically, few papers compare different architectures in the context of both accuracy and explainability.

This project aims to fill this gap by evaluating a custom CNN model with an imaging-processing transfer learning-based ResNet model for breast cancer detection using ultrasound images. Beyond experimental results concerning classification performance, the Grad-CAM is utilized to assess and make comparisons regarding model interpretability. The goal of this study is to foster accurate but also interpretable systems to get one step closer towards the establishment of clinically usable AI.

2.2 Technology Review

To ensure its success, the implementation of deep learning models for breast cancer detection depends on tools and frameworks as well as computer resources. As a viable solution to this problem, the project is built on Google Colab using popular libraries including TensorFlow and Keras for model building and training.

TensorFlow is an open-source deep learning framework developed by Google and used to build, train, and deploy any machine learning model. It offers a versatile framework for building complex neural networks in an efficient manner. Keras is a high-level API that comes bundled with TensorFlow, making it easy to build and train neural networks. Focusing on user experience allows for rapid prototyping and experimentation, which makes it ideal for academic projects.

The project implements both the custom CNN model as well as the transfer learning based ResNet Model using Keras. CNN architecture is built using Sequential API and pre-trained ResNet model is integrated using Functional API. It makes developments of custom models more efficient.

For this experiment we will take advantage of Google Colab as the development environment since it is free and provide computational resources. It offers free access to GPUs, allowing for faster training of deep learning models without requiring expensive local hardware. Collectively, this means that Colab can be highly useful and accessible as we build our code.

Data preprocessing and manipulation is done using image processing libraries such as OpenCV and NumPy. Specifically, OpenCV is used for the tasks of resizing images and normalizing it as well as displaying them and NumPy is utilized for numerical computation. Plt: For Plotting and Visualization (Training Curves, ROC Curve, Grad-CAM Output, etc.)

Standard performance metrics including accuracy, confusion matrix and ROC-AUC are calculated for model evaluation. By providing different aspects of the model performance, these metrics allow comparison between the various architectures.

These technologies have been chosen due to their prevalence in both the research and industry domains. TensorFlow and Keras provide the flexibility, scalability of running on CPU and GPU which have been discussed in this article, while Google Colab is a great cost-effective platform to run experiments. All in all, these tools help to make effective deep learning models for medical image analysis.

2.3 Summary

This chapter discussed the current literature and technologies of using deep learning for breast cancer detection. Several studies have shown that utilizing transfer learning methods with CNNs can lead to high classification accuracies [11], [12]. ResNet and EfficientNet Architecture improves its performance but increases complexity of Model.

Despite this, one major limitation evidenced within these studies arguably remains the non-interpretability of deep learning models. Accuracy is often prioritized at the expense of interpretability (which is vital in medical applications) in most studies.

This project compares CNN and ResNet models focussing on both performance and interpretability while visually explaining this model by means of Grad-CAM, addressing these gaps in the field. It also covered the tools and frameworks used for implementation, as well as why they were appropriate to use given that the techniques would be developed using deep learning.

All in all, this chapter lays the groundwork for the suggested methodology, emphasizing the requirement of interpretable and dependable AI systems in medicine.

Chapter 3: Implementation

3.1 System Overview / Pipeline

To build an automated breast cancer detection system based on deep learning methods, this work implements a pipeline of successive steps inbreeding the various intermediate results. Figure 1 is a complete workflow which includes data collection, preprocessing, model development/training/evaluation and interpretability evaluation. The objective is to develop a strong

and explainable model that can effectively classify ultrasound images as cancerous or non-cancerous.

The dataset is derived from a public source and follows specific categories at the beginning. Images are then preprocessed so that they have the same size and structure between them, allowing for effective model training. It justifies resizing, normalizing, and converting labels to prepare the problem for a binary classification task.

Once the data is preprocessed, we can build two different deep learning models. The first model is a custom CNN which directly extracts spatial features from the images. ResNet+Transfer learning: The second model uses a transfer learning approach, which means that the implementation of ResNet architecture uses pre-existing huminty as weights to learn better.

Using the prepared dataset both models are trained and evaluated through the common performance measures of classification, i.e., accuracy, confusion matrix & ROC-AUC. For better interpretability, Grad-CAM is employed to identify the parts of the images that affect the model's predictions. This enables us to compare how various models attend to relevant features.

3.2 Dataset Description

Dataset: The dataset used in this study is a publicly available breast ultrasound image dataset, which was obtained from Kaggle. Breast cancer screening commonly used ultrasound imaging because it is noninvasive, low-cost and yielder diagnostic to detect in dense breast tissue. The dataset consists of images labelled as normal, benign, and malignant (three classes).

Normal images indicate healthy breast tissue with no abnormalities visible. Benign images are associated with benign (non-cancerous) lesions which may need monitoring, but they are not fatal. Malignant visuals signify malignant tumors which must be a point of concern. These categories give a complete picture of real-world diagnosis.

This problem is simplified to a binary classification for the purpose of this study. The benign and malignant classes are combined into a single(class "cancer") class, while the normal class is treated as class non-cancer. Such a transformation is carried out in order to minimize class complexity, as the main goal in a clinical screening setting is separating between cancerous and non-cancerous cases.

The dataset shows variability in image quality, contrast, and tumor appearances mimicking real-life challenges in ultrasound imaging. Its variations make it apposite etc testing robustness of deep

learning models. Exploratory data analysis (EDA) is performed on the dataset before training a model to better grasp its distribution, features and limitations.

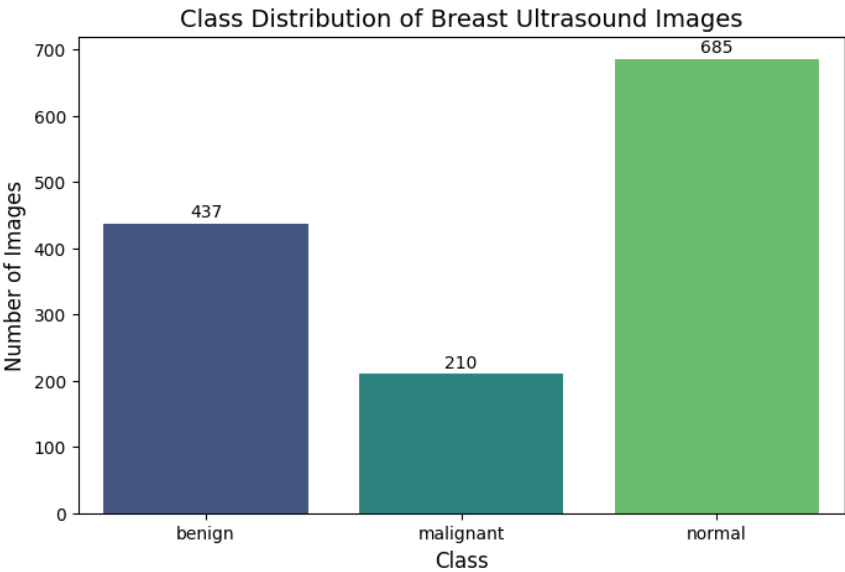
3.3 Exploratory Data Analysis (EDA)

Table 1: Dataset Distribution

Class	Number of Images	Percentage (%)
Normal	685	51.4%
Benign	437	32.8%
Malignant	210	15.8%
Total	1332	100%

The Exploratory Data Analysis (EDA) phase is essential in preparing the data before an analytical model can be built, as it allows us to get a better idea about the properties, structure and quality of the dataset that we are currently working with. This study performed in-depth exploratory data analysis (EDA) on the data set of breast ultrasound images to observe patterns, variations, and challenges that are likely to contribute towards the model performance. The study analyzes class distribution, image features, dimensional characteristics, and intensity differences.

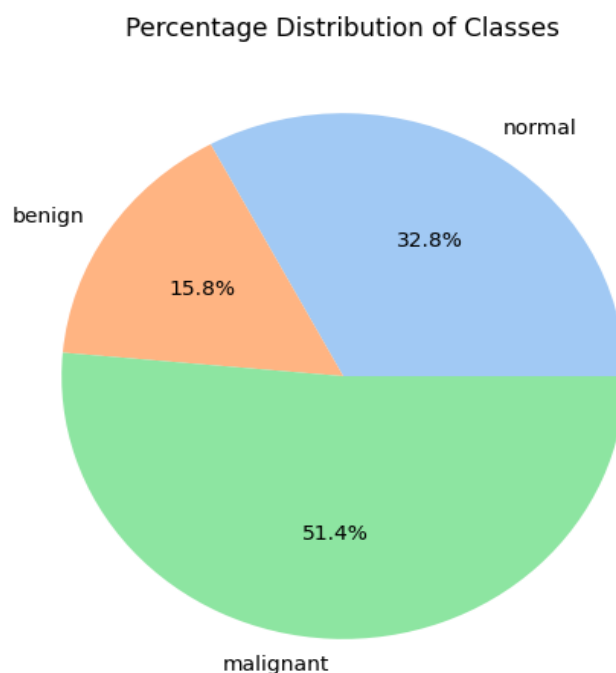
Figure 3.1: Distribution of breast ultrasound images per class



The third part will show the class distribution of dataset and we can clearly see that images are divided into 3 classes or tumour types which is Benign, Malignant and Normal. It comprises about 437 benign images, 210 malignant images and 685 normal images. It can be seen that there is a stark class imbalance, where the representatives of normal class is the highest. Benign comes second in rank and malignant case remains underrepresented.

The imbalance is particularly important as the deep learning approach can be biased towards majority classes, thus resulting in an inadequate performance for malignant class detection, which is clinically more relevant. This issue was solved by changing the classification problem into binary with benign and malignant classes being merged under a single “cancer” category. This conversion aids in balancing the dataset and ensures that the model is tailored to practical clinical site screening goals where identifying cancerous from non-cancerous cases takes precedence.

Figure 3.2: Distribution of the classes in percentage

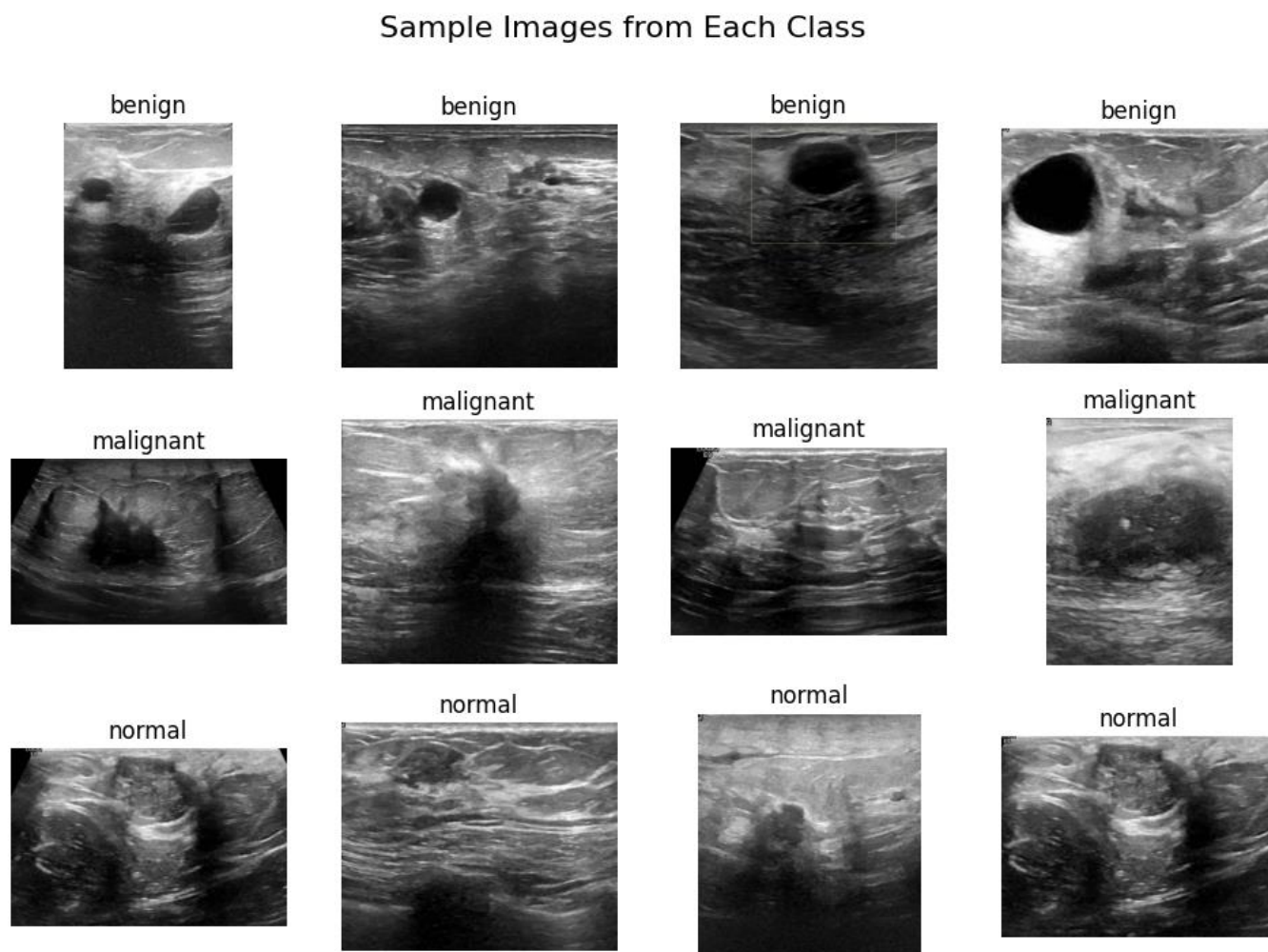


The relative percentage distribution also reflects the skewed nature of the dataset. For the complete dataset, malignant cases comprise around 15.8%, benign cases 32.8% and normal both about 51.4%. The small ratio of malignant instances showcases the challenges to detect cancer cases.

If the class is unbalanced, from a Machine Learning point of view that can lead to biased learning of the model wherein the dominant class is prioritized. This provides a dual use case for binary classification, and highlights the significance of evaluation metrics like ROC-AUC and confusion

matrix that allow you to measure not only how accurate the model is but also take into consideration false predictions.

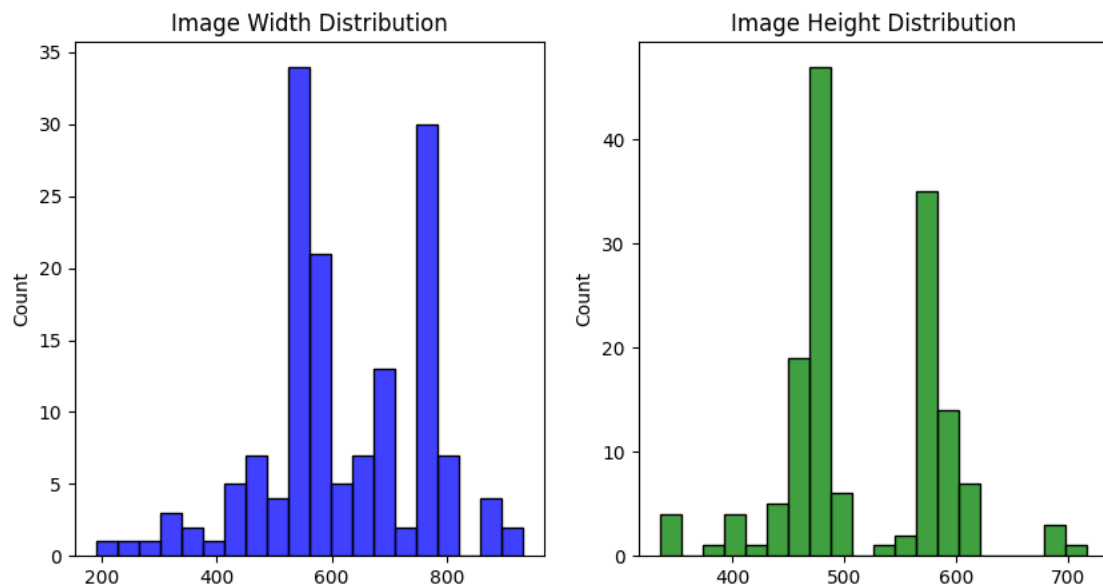
Figure 3.3: Examples of Images Sampled from Each Class



Looking at some sample images for every class, gives us a very good feel about what type of data we are dealing with. Benign shapes commonly display clear and smooth margins, whereas malignant forms are frequently more irregular with jumbled borders and heterogeneous appearances. Normal images exhibit consistent tissue structures without evident abnormalities.

Ultrasound images, however, are inherently noisy with low contrast and have speckle artifacts so that visually distinguishing between classes is a challenge. Benign and malignant lesions are similar in some cases, complicating the classification task. This observation speaks to the power of deep learning models that can find subtle patterns not easily recognized by humans.

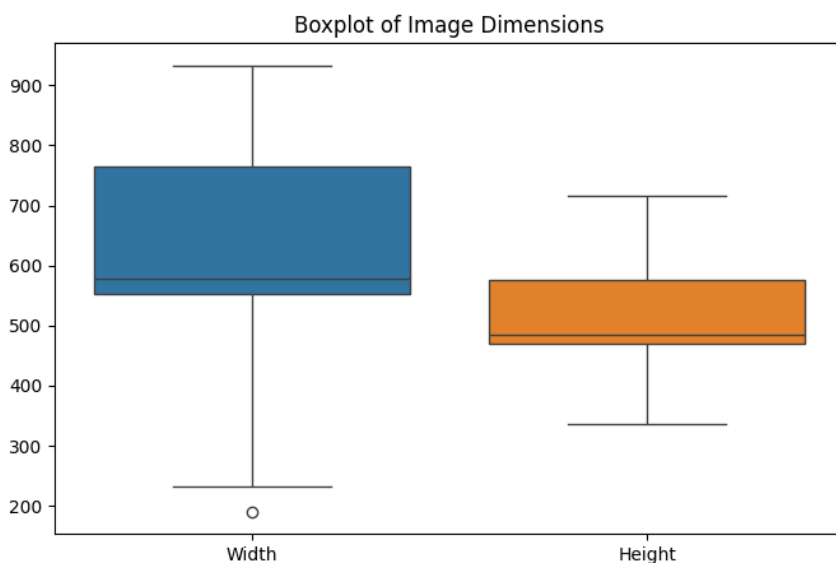
Figure 3.4: Image Width and Height Distribution



The distribution of image size also has a lot of variance along both width and height. Image widths are around 200-900 pixels and heights are between 350-700. This inconsistency suggests that the dataset is not uniformly resolved.

This variability can lead to poor model training since neural networks expect a fixed dimensionality in inputs. Thus, during preprocessing same size is applied on all images by resizing them to 224×224 pixels. This helps in getting more consistent data, while also allowing the deep learning models to efficiently extract features.

Figure 3.5: Boxplot of the image dimensions



Additionally, to further examine the distribution of image dimensions, we perform a boxplot analysis. It can be seen that there is a bigger spread in the width, as well as outliers, whereas more height measurements are around the median.

Here, the presence of outliers indicates that a few images have

very different proportions from the rest. If not handled properly, these variations can create noise in the model. This can be addressed to certain extent by resizing and normalization processes.

Figure 3.6: Image Width by Height Scatter Plot

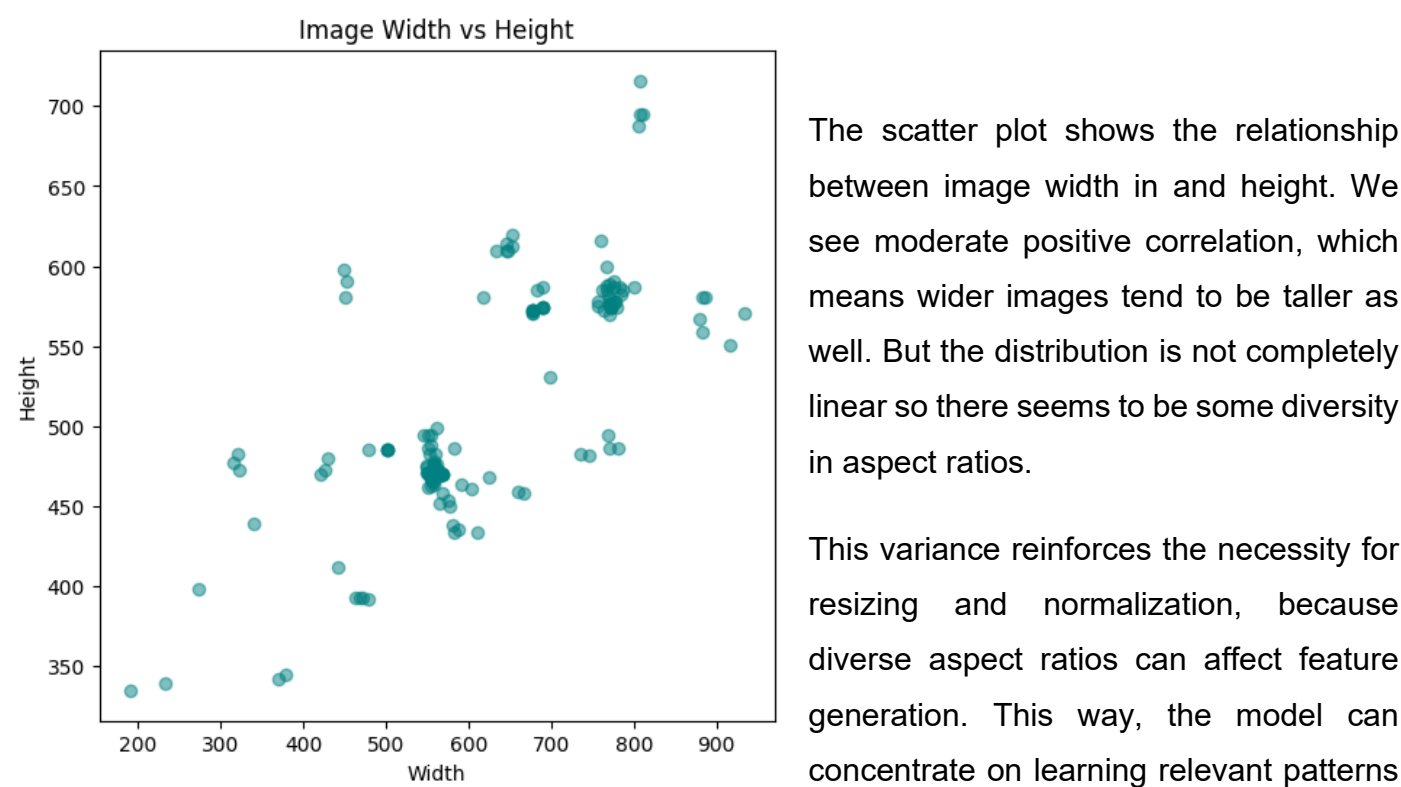
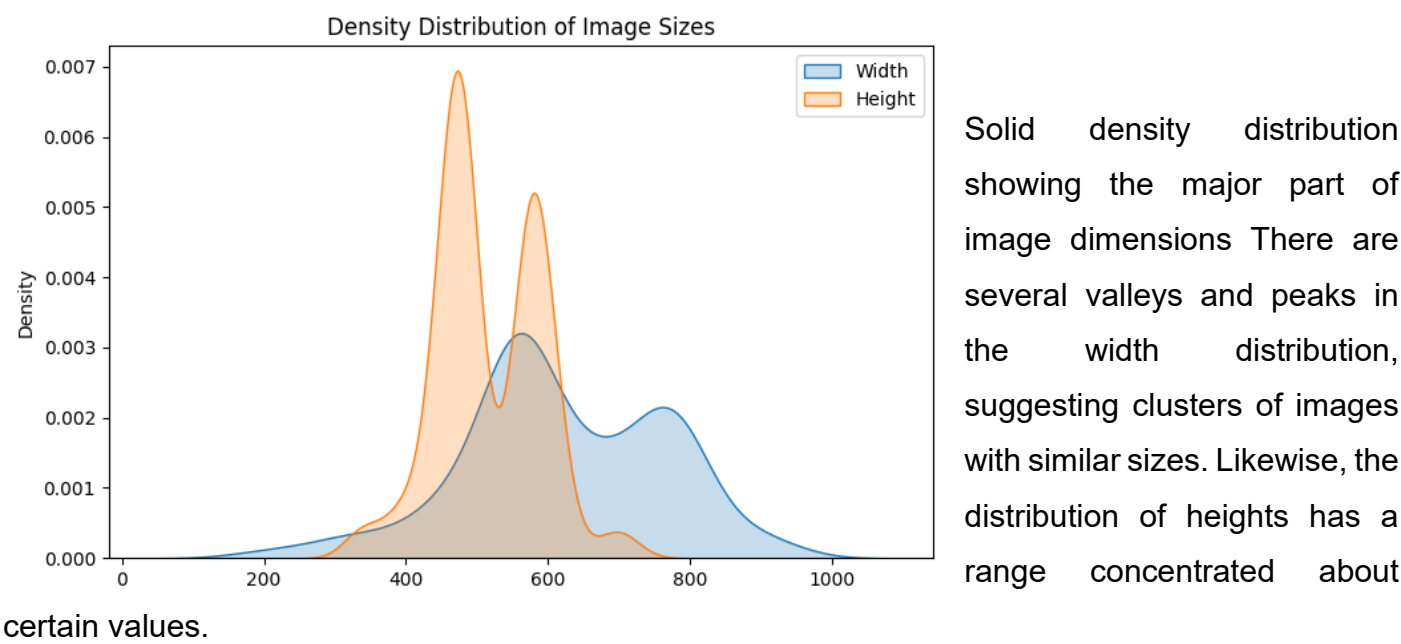


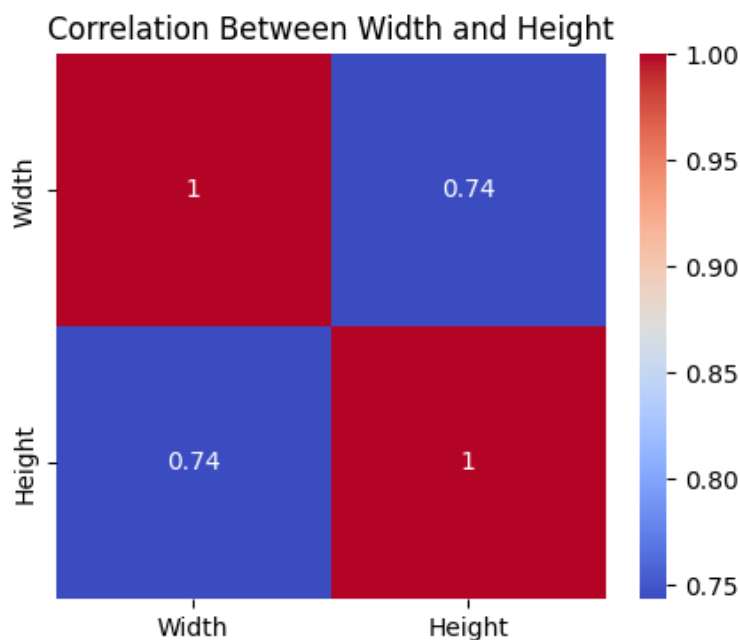
Figure 3.7: Image Size Density Distribution



These patterns indicate that the dataset comprises clusters of images taken under remarkably similar conditions or with comparable equipment configuration. This distribution is important to

understand in order to ensure that the preprocessing steps are designed in accordance with such variations.

Figure 3.8: Net Width vs Height Correlation

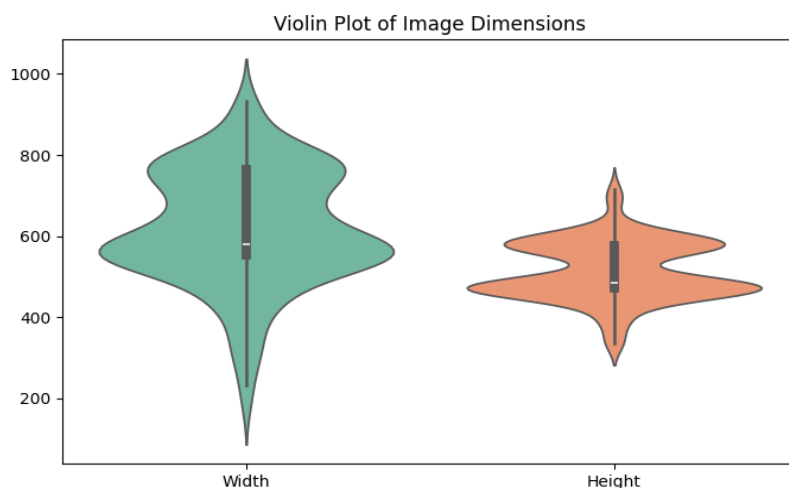


The heatmap shows a strong positive correlation of about 0.74 between image width and height. This implies that when one dimension goes up, the other tends to go up as well.

While this relationship is expected, it also verifies that the dataset has a reasonable scaling across dimensions. But the correlation is not perfect, which further supports why we need resizing to make

sure all of the input settings are consistent with each other.

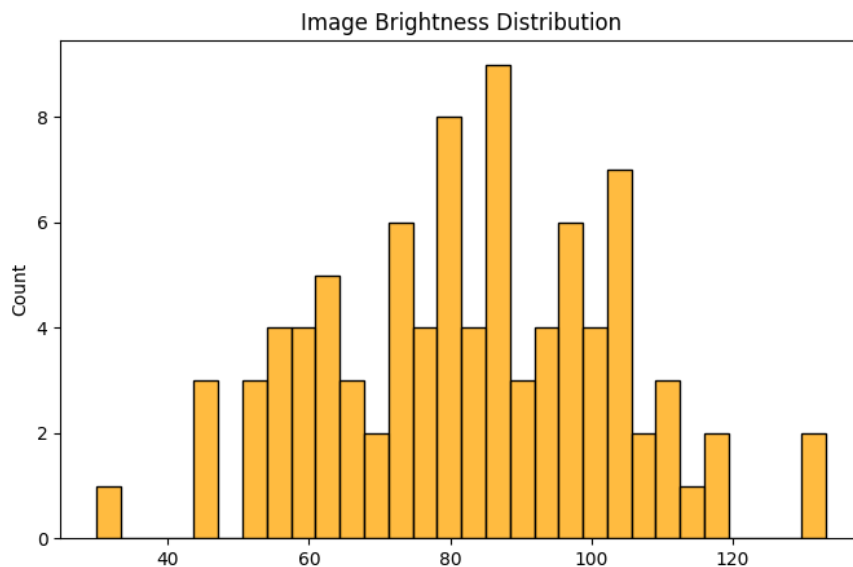
Figure 3.9: Violin Plot of Image Dimensions



The violin plot shows us a more elaborate representation of the image dimensions distribution and density. Width distribution is more dispersed with several local peaks, whereas height distribution is more compact.

The visualization shown above clearly shows that width variation is greater than height variation, given that this affects how features are pulled out of convolutional layers. This difference does not spoil performance, if we keep input dimensions uniform.

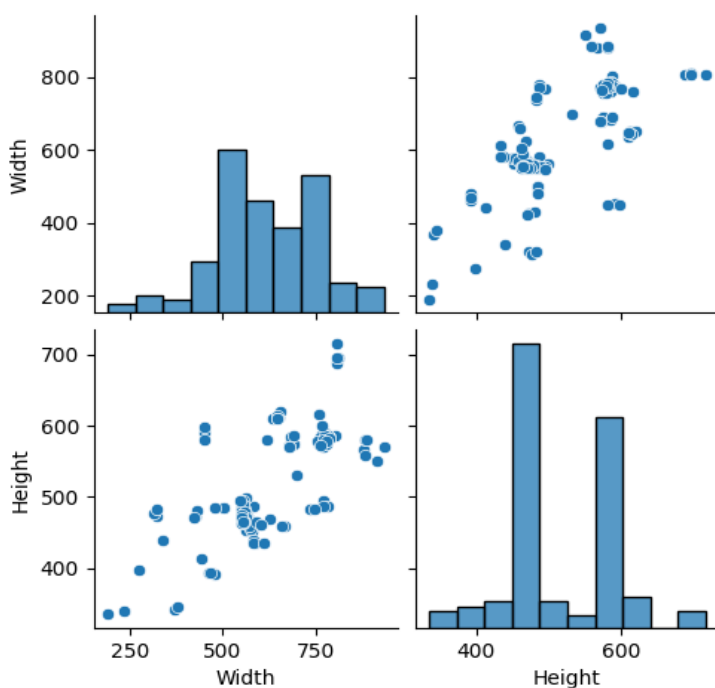
Figure 3.10: Image Brightness Distribution



This brightens up the brightness distribution for different pixels on the image. The majority of the images have intensity between mid to high ranges but there are some fallen values in lower intensities as well.

Variations such as these are common in ultrasound imaging, due to different acquisition settings and tissue types. Normalized pixel values between 0 and 1 are required to guarantee convergence during the learning phase.

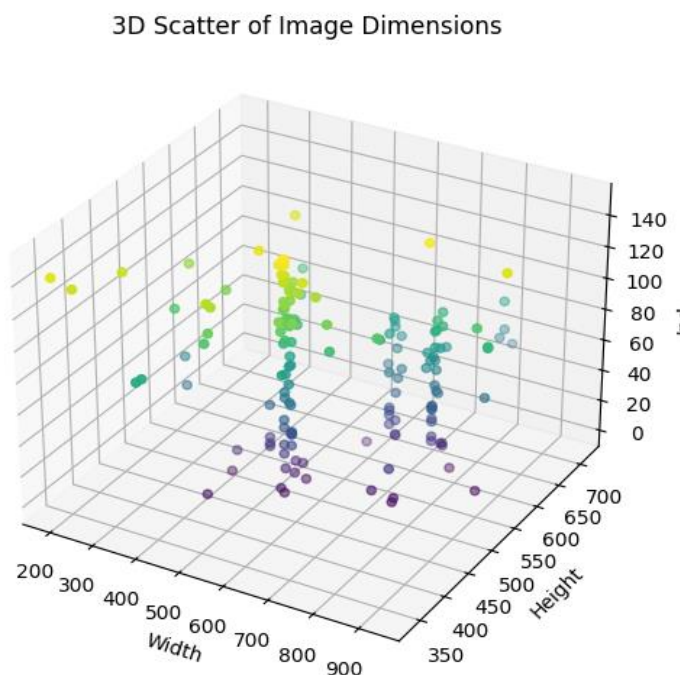
Figure 3.11 Pair Plot of Image Dimensions



The pair plot allows you to see the distributions of a variable and its relationship with other variables. This fits the same patterns seen in earlier plots, including image size clustering and moderate dimensional correlations.

These points reinforce the need for having a standard and consistent shape of input to the DNN algorithm = Image Preprocessing at this stage.

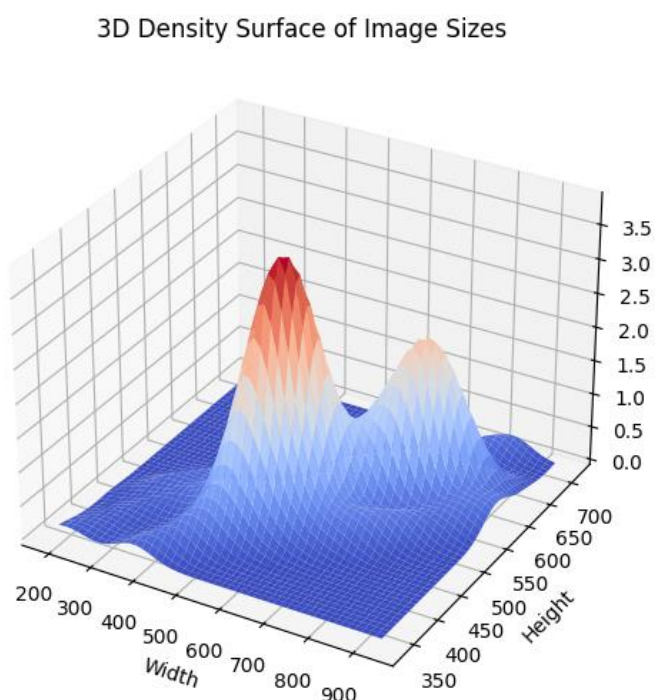
Figure 3.12: 3D Scatter Plot of Image Dimensions



An extended visualization of image dimensions, where the additional dimension is intensity/index in a 3D scatter plot. Clusters of images reveal such clustering patterns, showing which images share common properties.

The fact that these are clustered indicates that there can be sub-sets of this dataset where images were taken under the same conditions, which is important when it comes to model generalisation.

Figure 3.13: 3D Density Surface of Sizes of Images



The 3D density view shows the concentration process of image features in a continuous space. If you think of this as a saddle, the higher peaks are areas encompassing most of your image sizes and flatter areas represent less common dimensions.

This visualization helps to confirm that the dataset is not uniformly distributed with regard to image size, as discussed in Step 1.2. This also reinforces that well-implemented preprocessing steps such as resizing and

normalization are critical aspects of the process

Summary of EDA Findings

The exploratory analysis yields a few insights worth noting. For one, the dataset is unbalanced with relatively few malignant cases. The second reason is the high variance of sizes and intensities in

images. Third, ultrasound images include noise and low interference which makes the classification difficult.

These insights guided the design of both our preprocessing and modeling pipeline. The problem was approached through binary classification to reconstitute class imbalance and used resizing and normalization techniques for inputting data. In summary, the EDA offers excellent guidance that sets the stage for creating effective and interpretable deep learning architectures.

3.4 Data Preprocessing

Due to the variability and noise in data, especially in medical imaging applications, preprocessing of input is an essential phase while developing the deep learning model. In this study, we developed a structured preprocessing pipeline to standardize the breast ultrasound images and improve their trainability in the model.

During exploratory data analysis, one of the first challenges identified was that individual image dimensions were not consistent. Images varied in width and height that were not compatible with convolutional neural networks, which expect fixed-size inputs. To solve this problem, all the images were resized to a uniform resolution of 224×224 pixels. We use this size since it is a widely used size for many deep learning architectures both as user defined CNNs and real-world pre-trained models like ResNet.

To standardize the intensity values of the images, pixel normalization was also done along with resizing. Because acquisition settings and the properties of different tissues are different, ultrasound images may have brightness and contrast variations. To reduce this variability, the pixel values were normalized to a interval between 0 and 1 through dividing by 255. This normalization gives better numerical stability during training, leading to a better convergence of the model.

Lastly, one more important preprocessing action was label transformation. The dataset originally comprised of three classes: normal, benign and malignant. However, because of class imbalance and the clinical goal of differentiating cancerous from non-cancerous cases, the problem was reformulated as a binary classification task. In the transform, benign and malignant images collapse to a single “cancer” class while normal images were labeled as “non-cancer.” This simplification helps in balancing the dataset, which is more appropriate since we are dealing with a realistic early cancer screening problem where most important thing is to detect cancer at an early stage.

An 80:20 split with respect to the dataset for training and testing subsets was used to ensure strong evaluation of models. Stratified sampling was used through this process to keep the unique class

distribution in both subsets and to avoid being biased towards a particular class. This is done to ensure the trained model gets evaluated on a representative sample of the dataset.

Also, the dataset was transferred to NumPy array for fast handling in TensorFlow language based models. Training also employed batch-wise processing to balance efficiency and memory.

While simple data augmentation approaches advanced techniques (rotation, flipping or zooming) can improve the generalization power of models; they were not used extensively in this study due to the need to preserve the original features of medical images. This prevents distortion of clinically relevant features during training.

In general, the preprocessing pipeline successfully tackles the issues discovered in exploratory analysis: variation between image size, intensity and class imbalance. This step provides the momentous basal required for accurate treatment and reliable breast cancer classification by standardizing the dataset according to the needs of deep learning models.

3.5 CNN Model Development

This study aimed to develop a baseline CNN model for binary classification of breast ultrasound images. Due to their capability of learning hierarchical spatial features automatically, CNNs are extensively used for medical image analysis applications that involve pattern recognition such as tumor detection and examination of tissue abnormalities.

CNN Model Architecture

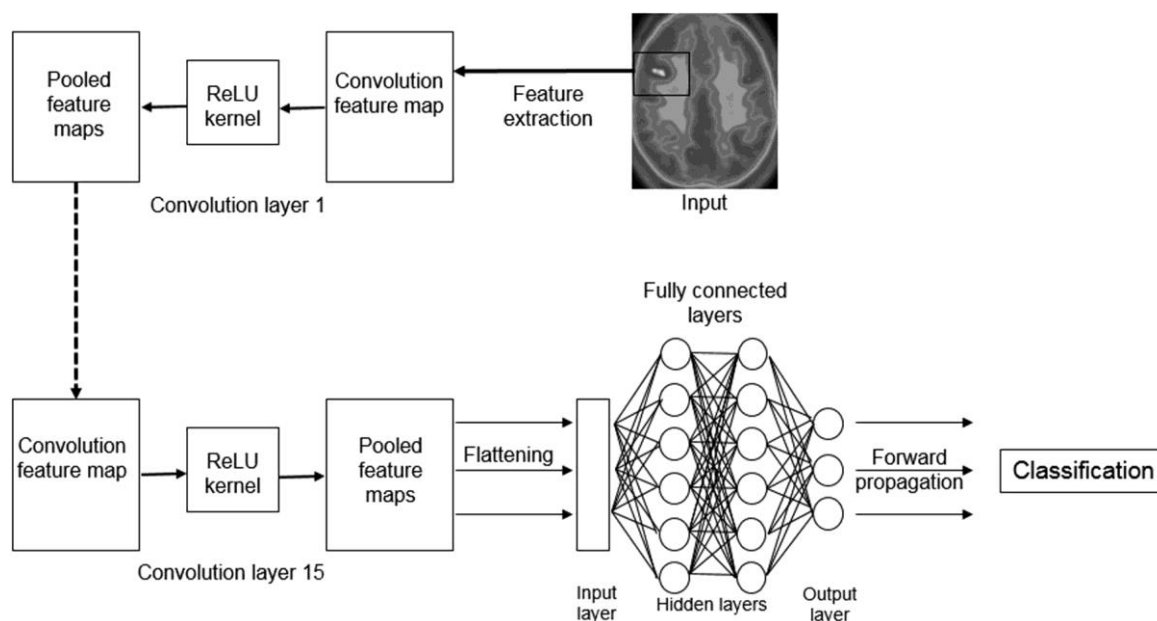


Image source : [Intelligent Imaging: Anatomy of Machine Learning and Deep Learning](#)

Input Image - Convolution Layer (contains Kernel) - ReLU Activation Function - Pooling Layer Flatten Output of Pooling - Fully Connected Model Input: A $224 \times 224 \times 3$ Preprocessed Image The first convolutional layer contains 32 filters of kernel size 3×3 with ReLU activation function. It is the first layer to encode low-level features which include edges and textures. Next, a max-pooling layer with pool size 2×2 is applied to make the feature frame smaller How does the number of pixels and reduce the computational complexity.

The network slowly increases its depth to learn more sophisticated features. Another max-pooling layer is then followed by a second convolutional layer with 64 filters. The same process was repeated for a third convolutional layer with 128 filters to learn even more higher-level features such as shapes and structures of the lesions. That is where this hierarchical feature extraction plays the key role in differentiating between cancer and healthy regions of ultrasound images.

After the convolutional blocks, a flattening layer holds the feature maps as they are flattened into a 1D vector. Next, we add a fully-connected (dense) layer with 128 neurons and ReLU activation to learn complex combinations of extracted features. To prevent overfitting, we introduce `tf.keras.layers.Dropout`. In our case Dropout at 0.5 means we're telling the algorithm to randomly disable neurons (and their connections) during training time by a rate of 0.5

The output layer is typically a single neuron with a sigmoid activation function, which outputs the probability that an image belongs to the cancer class. This configuration is ideal for binary classification tasks.

The model was compiled with the Adam optimizer. Binary cross-entropy was chosen as the loss function because it works well for problems of binary classification. Accuracy was our main metric to assess the performance of the model.

We trained on 100 epochs with a batch size of 32. Stable convergence during training with high training accuracy and consistent validation performance were observed. Our CNN model gave very good accuracy of about 94%, thus proving that CNN is able to take necessary features from ultrasound images.

Beyond classification performance, interpretability was integrated through Grad-CAM visualization. This helped in making the model decision process interpretable by highlighting regions of interest in images. The CNN-based Grad-CAM result presented significantly localized activation, indicating the model has learned clinically relevant features by concentrating on tumor regions.

In summary, this is a simple yet effective custom CNN model capable of giving an easy baseline for breast cancer classification with the very high accuracy. Furthermore, its ability to give interpretable results makes it even more appropriate for medical related applications.

3.6 ResNet Model Development

To further improve feature extraction and model generalization, a transfer learning approach using the Residual Network (ResNet) architecture was also implemented alongside the custom CNN model. ResNet, also known as residual network is a deep convolutional neural network which uses shortcut or residual connections to resolve the problem of vanishing gradient and helps in training very deep architectures efficiently.

ResNet Model Architecture

Keras ResNet⁵⁰

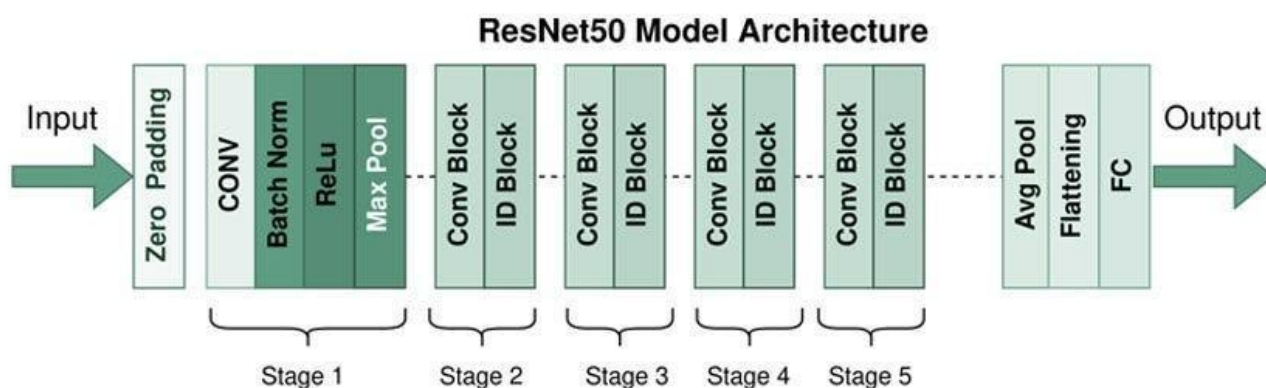


Image Source: [Residual networks architecture](#)

In this study, the ResNet model pre-trained was used as a base network. Weights from an ImageNet trained model were used to initialize the model so it can capitalize on the building blocks of learnt features (such as edges, textures and complex visual patterns) already present in this model. As a result, training time is largely reduced and performance improves notably when we can only work with small sets of medical data.

By tuning the last layers of the network to fit the breast cancer classification task, we adapted this architecture for our data. We removed the original dense layers of ResNet and added a custom classification head owing to binary classification. The output includes a Global Average Pooling layer, used to reduce the spatial dimension of feature maps followed by a batch normalization layer

used for stabilizing learning. We added a dense layer using ReLU as the activation function to learn task-specific features and used dropout layers to remove noise.

A single neuron with a sigmoid activation function in the final output layer provides a cancer detection probability score. This is in line with a binary classification task, where we want to classify the input as being cancerous or non-cancerous.

Since partial fine-tuning of the ResNet model allows it to learn effectively, First, the layers were frozen that had been learned from the pre-trained model in order to save the features they learnt, then only trained newly added layers. Then, from this base model, certain deeper layers were unfrozen for fine-tuning so the model could adapt better to the ultrasound imaging domain.

We compile the model with binary cross-entropy loss function using Adam optimizer. In order to have a fair comparison, training was performed on the same dataset and with the same preprocessing pipeline as for the CNN model. The accuracy of the model was ~91%, which is marginally lower than the custom CNN model.

The ResNet model was capable of extracting features very well but had some flaws. Compared to the CNN model, the Grad-CAM visualizations produced from ResNet model were more diffuse and less localized. The activations were not focused on the tumor region, but relatively diffuse over larger areas of the image. It implies that the model might be leveraging macroscale context instead of fine-grained tumor-specific information.

Training a ResNet with more layers also requires increased computational resources and longer training times. The horses approach has great results because transfer learning give a strong starting point but it won't always perform better than a simpler model, especially on a small dataset which is rather domain-specific.

Here, we have it shown how transfer learning works well in the case of medical image classification using ResNet model. Yet in this study, the custom CNN model was superior to ResNet with respect to both accuracy and interpretability. This underlines a need for choosing the appropriate model architecture with respect to both dataset characteristics and application requirements.

3.7 Model Training Strategy

Properly defining the training strategy is key to obtaining good performance in deep learning models, and this becomes critical in tasks with class imbalance and variability from an image data perspective -- for example, medical images; as can be found in our previous article.. The CNN and

ResNet were trained using a consistent approach to ensure that performance could be compared fairly, without introducing bias due to different methods of training.

An 80:20 split was used to divide the dataset into respective training and testing sets with stratified sampling to maintain class distribution across both. This prevents training and evaluation on the same data, helping to mitigate the risk of biased performance. Model parameters were learned from the training set and final evaluation was carried out using a separate testing set.

Both models were trained with batch size as 32, which provides an optimal point for the training time and stable gradient updates. Training was done for several epochs to enable the models to adjust their learnt features iteratively. To speed up convergence and improve stability during training we used the Adam optimizer. This optimization mechanism works best for complex image-based tasks.

For both models, the binary cross-entropy was utilized as a loss function to fit both the model against each other due to their binary classification problem. This loss function quantifies the difference between the predicted probabilities and the true labels, encouraging learning that minimizes classification mistakes. We primarily used accuracy for model evaluation during training, but we also looked into other metrics like confusion matrix and ROC-AUC for a more holistic view of the model behavior.

Validation data was used to monitor model performance and prevent overfitting, with a validation check at the end of each epoch. Convergence behavior was assessed by analyzing training and validation loss and accuracy curves. Although the ResNet model was slightly more volatile than the CNN model - likely due to its greater complexity—the pattern was clear; the CNN model exhibited stable training and steadily increasing validation accuracy.

A major issue in deep learning is overfitting, particularly when dealing with small datasets. In order to combat this, both models included regularization techniques like dropout. It encourages the network to learn more generalized features by randomly "dropping out" a subset of neurons (along with their connections) during training. And, principles of early stopping allowed us to consider the fact that training must stop when results are optimal and should not extend further.

The training results show that CNN has a higher accuracy and more stable convergence than ResNet. It is also worth noting that relatively straight forward architecture can do better compared to overfitted architectures on smaller domain-specific datasets.

To sum up, the training strategy while adhering to chronological order prevented information leakage and maintained fair comparison and consistent model development which helped obtaining reliable evaluation. Through thoughtful hyperparameter selection, optimization approaches, and validation

strategies used in the study, a solid base for precise breast cancer classification and significant performance assessment is laid.

Chapter 4: Evaluation and Results

4.1 Performance Metrics Overview

In this study, we evaluate the performance of proposed models with different metrics to cover a detailed analysis of classification performance. Although providing an overall evaluation of performance, accuracy is meaningless for medical diagnosis tasks where both class imbalance and misclassification risk have to be taken carefully into account. Hence, other metrics like precision, recall, F1-score, confusion matrix and ROC-AUC are used.

Precision is the ratio of true positives to all predicted positives, while recall (sensitivity) quantifies how well the model predicts actual cases of cancer. F1-score is a harmonic mean of precision and recall, which becomes especially useful when working with imbalanced data. Recall is particularly important in medical applications, where missing a cancer case (a false negative) can be detrimental.

Next, confusion matrices are employed to visualize the misclassifications and ROC curves ascertain how well the model separates classes at different thresholds. Grad-CAM visualization is also used to interpret the decisions by the model and to ensure that predictions were made on clinically relevant areas.

Together, these metrics offer a comprehensive set of tools to assess and contrast the performance of CNN and ResNet models.

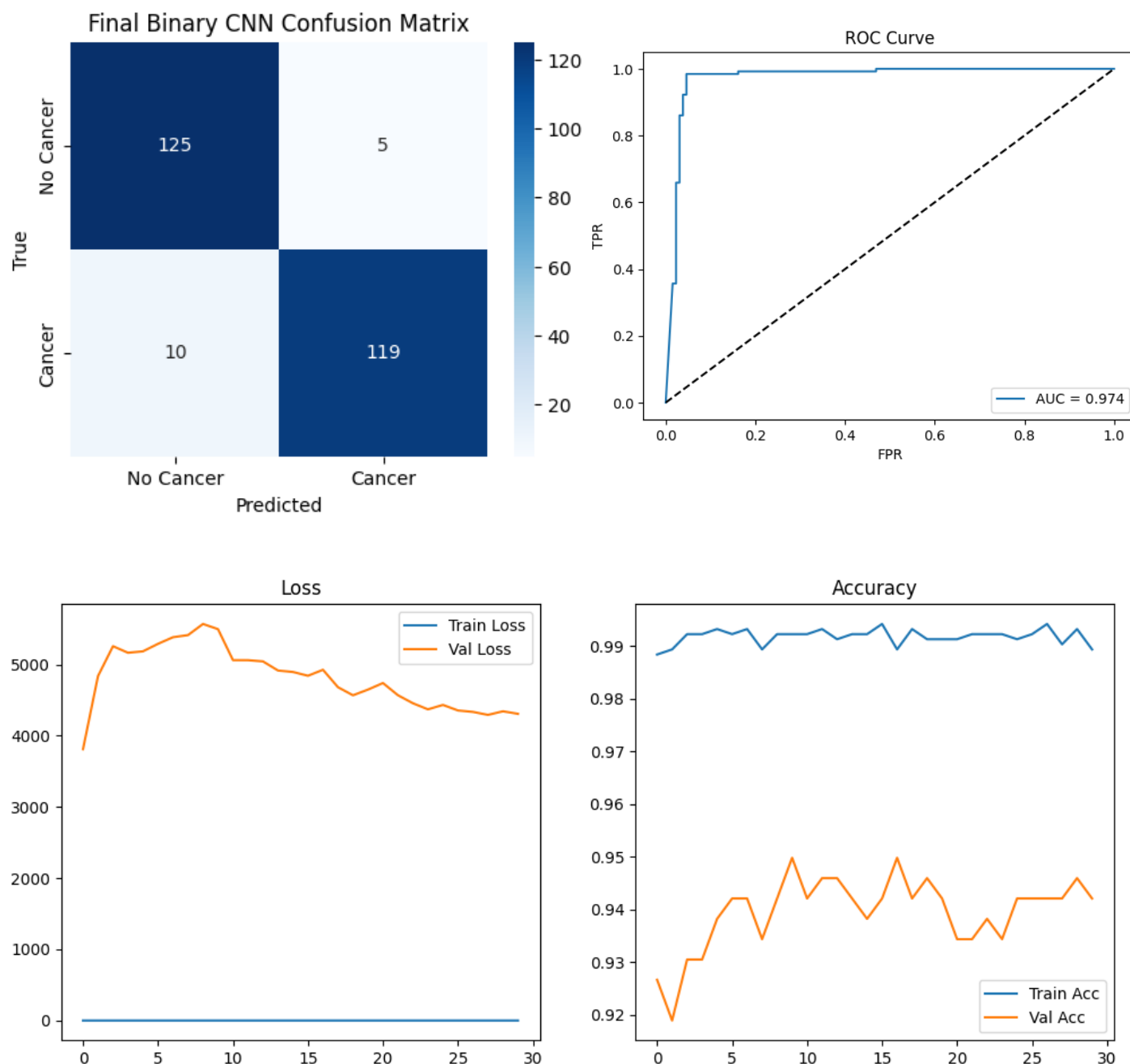
4.2 CNN Model Evaluation

The CNN model provides stable and good performance in all evaluation metrics. Overall accuracy on the ground truth is 94% and it seems that performance among both positive/negative class (0/1) are balanced.

	precision	recall	f1-score	support
0	0.93	0.96	0.94	130
1	0.96	0.92	0.94	129
accuracy			0.94	259
macro avg	0.94	0.94	0.94	259
weighted avg	0.94	0.94	0.94	259

The CNN is also able to obtain a precision of 0.93 and recall of 0.96 for the non-cancer class, suggesting that healthy cases are identified as well while maintaining a low false positive rate. That means for cancer, precision is 0.96 recall is 0.92 This indicates that the model makes very few false positive errors but sometimes misses true positives (i.e., a healthy patient getting diagnosed to have cancer).

Figure : CNN Classification Report, Confusion Matrix, ROC Curve, Loss & Accuracy Graph



This is also illustrated by the confusion matrix. Out of 125 non-cancer cases it predicts correctly, and 119 cancer cases it classifies correctly. It also shows that the CNN is able to balance sensitivity and specificity well, which are crucial for medical diagnosis.

The ROC curve of CNN model indicates the excellent performance with an AUC value of 0.974. The closer to the top-left corner of the plot, the better a classifier performs in terms of high true positive rates and low false positive rates at various threshold values.

Learning behaviour is stable as seen by the training and validation curves Training accuracy is consistently high, validation accuracy increases and stabilizes slightly above the 94% region. There are some fluctuations in the validation loss, but the model starts without severe overfitting.

The CNN model produces general performance and reliable robustness suitable for breast cancer detection tasks.

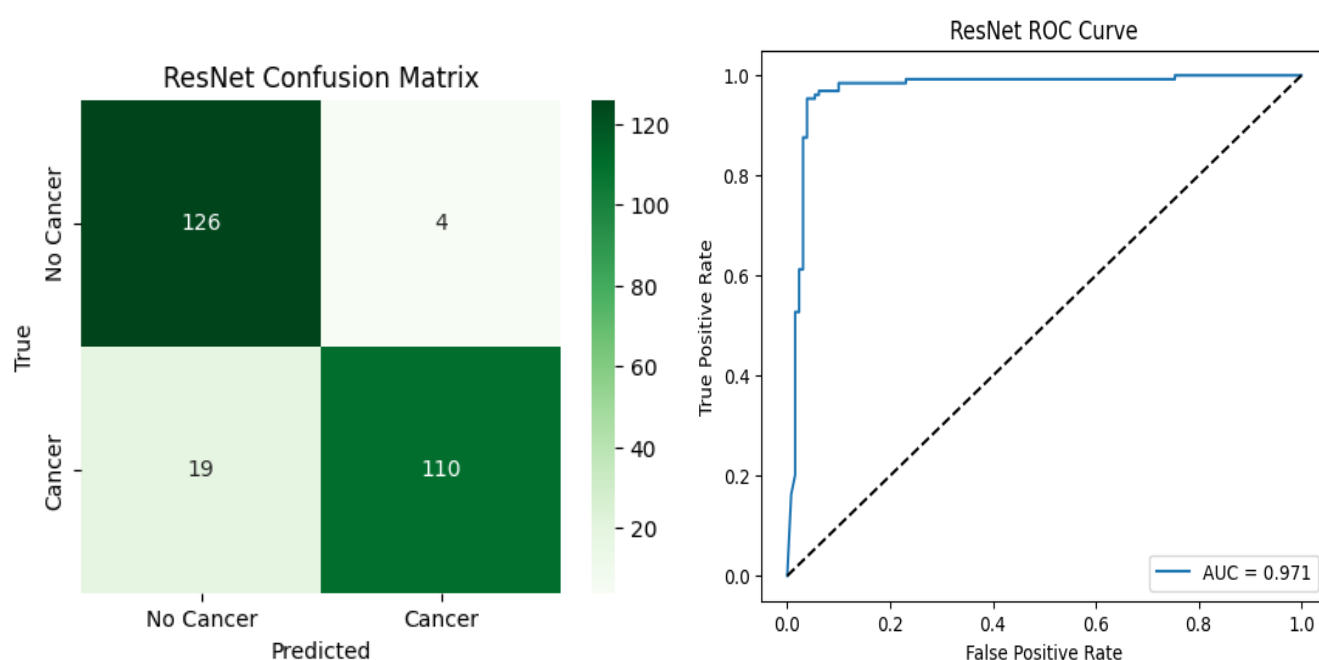
4.3 ResNet Model Evaluation

	precision	recall	f1-score	support
0	0.87	0.97	0.92	130
1	0.96	0.85	0.91	129
accuracy			0.91	259
macro avg	0.92	0.91	0.91	259
weighted avg	0.92	0.91	0.91	259

While the ResNet model also performs well, it is slightly lower than the CNN model. As we can see in the classification report, overall accuracy here is

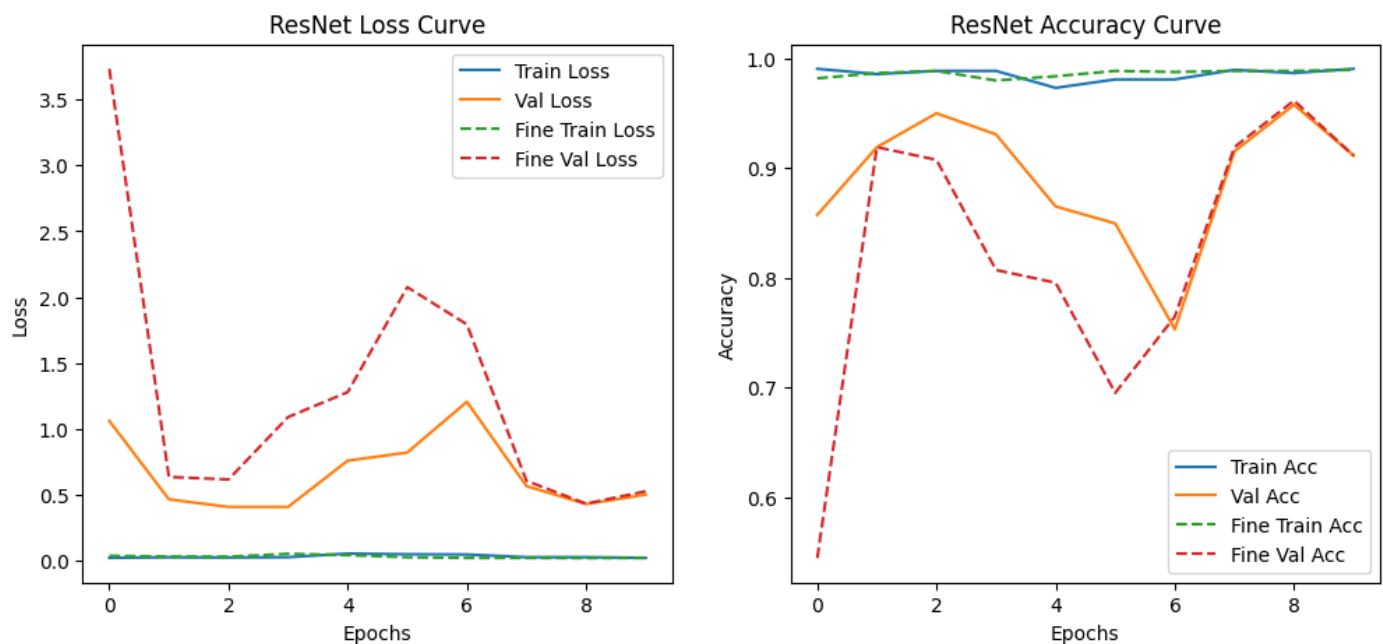
91% which indicates the model is good but not very consistent.

Figure: ResNet Confusion Matrix, ROC Curve, Loss & Accuracy Graph, Precision-Recall Curve



The model has a precision of 0.87 and recall of 0.97 for the non-cancer class. This means that the model is very precise in detecting healthy individuals with no signs of cancer compared to the other methods, but creates more false positives in comparison. The precision of cancer class is 0.96 and recall is 0.85, indicating the model misses a higher number of cancer cases.

This limitation is highlighted by the confusion matrix. Although it correctly predicts 126 non-cancer cases, it generates 19 false cancer predictions which poses a serious problem in medical domain.



The ROC curve displayed good discriminating capability with an AUC value of 0.971. While close to the performance of CNN, actually lower proving that there is marginal better performance given by CNN.

We can see from the training and validation curves that the ResNet model exhibits a higher variability while training. The validation loss shows many spikes indicating instability, and possible overfitting. This might be because of the complex function and the small dataset.

In summary, considering that ResNet and CNN model apply to transfer learning for boosting the performance of detection accuracy, the overall result demonstrates that ResNet model proves less stable thus not much reliable comparing with CNN model.

4.4 Grad-CAM Analysis (Interpretability)

Medical applications of deep learning models raise challenges in interpretability. Grad-CAM allows the visualizations of parts of an image that influence a model's predictions, creating interpretability in model performance.

Figure: CNN Grad-CAM (Original , Heatmap , Overlay)

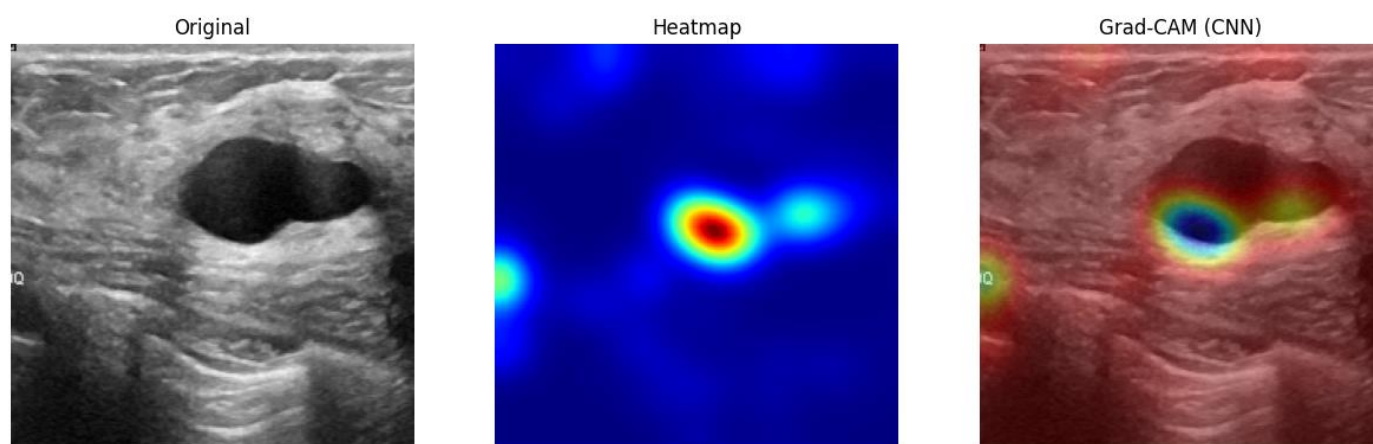
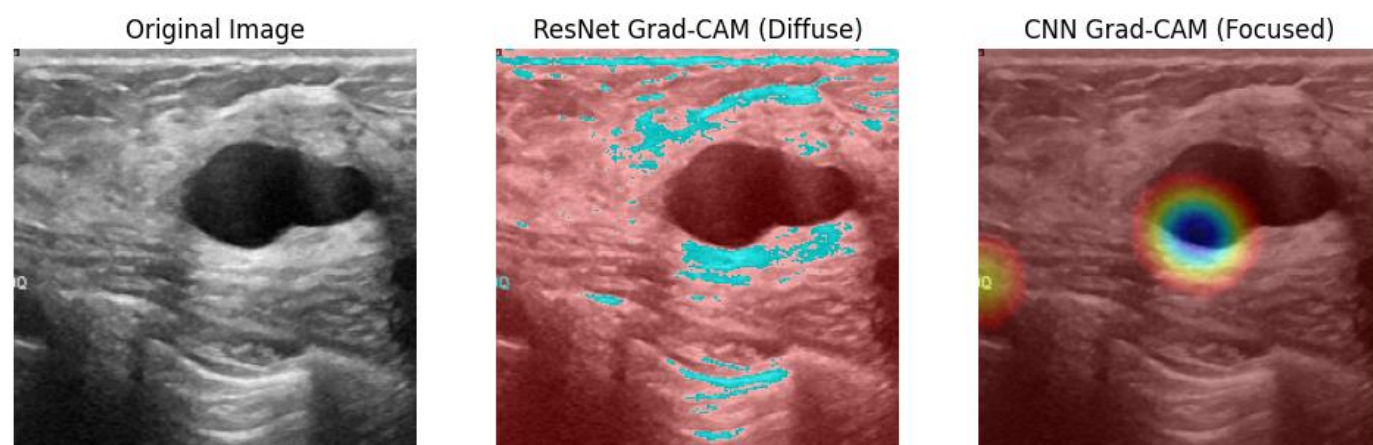


Figure Grad-CAM results From the CNN show a very localized activation only in the tumor region. These are heatmaps which show where the lesion is, and this reflects that the model is learning features of cancer. The model becomes more trustworthy and reliable due to this selective attention.

Figure: CNN vs ResNet Grad-CAM Comparison



The ResNet Grad-CAM results seem diffuse and scattered around the image, in contrast. The model pays more attention... This indicated that the ResNet model based on less specific features and poor performance.

Grad-CAM results show the difference which is a strength of CNN model. As it achieves higher accuracy, it also offers more interpretable and clinically meaningful prediction results.

4.5 Comparative Analysis

Table 2: Model Performance Comparison

Metric	CNN Model	ResNet Model
Accuracy	94%	91%
Precision	0.96	0.96
Recall	0.92	0.85
F1-Score	0.94	0.91
AUC-ROC	0.974	0.971

Thus, when comparing both models, CNN model showed better performing results than ResNet. The CNN performed better, achieving 94% vs 91% accuracy and a higher sensitivity (recall) for cancer detection, which is crucial in medical diagnosis.

As shown in the confusion matrix analysis, CNN yields a lower number of false-negative than ResNet confirming that it is more reliable for cancer cases. Moreover, CNN demonstrates more consistent behaviour during training compared to ResNet that exhibits fluctuations of performance on validation set.

From the perspective of interpretability, the CNN does yield more focused Grad-CAM visualizations which target on tumor clearly. On the other hand, Diffuse heatmaps generated by ResNet that shows no anti-aliasing of learned features.

As such, while ResNet performs well with transfer learning, it may be overfitting on the smaller dataset given its complexity. The CNN, more task-oriented and simple, generalises better and performs well.

4.6 Summary of Results

Evaluation results demonstrate that both models are capable of performing effective classification of breast cancer. The CNN model performs better than the ResNet model on all evaluation metrics, however.

In comparison, we find that the CNN has higher accuracy and recall values and a more stable training behavior. It additionally provides more interpretable results with focused Grad-CAM visualizations. The ResNet model, as a powerful but challenging entity for Recall and Interpretability.

These results emphasize the need of defining a suitable model architecture in accordance with dataset characteristics. The custom CNN model is found to be more suitable for breast cancer detection from ultrasound images in this study.

Chapter 5: Conclusion

In this study, we proposed a new, fully deep learning-based model for the automated detection of breast cancer using ultrasound images. We aim to establish and validate high performance classifiers that would be able to distinguish cancerous from none-cancer cases with interpretable models of decisions. In order to attain this, two different architectures were developed: a custom ConvNet and a Residual Net using transfer learning.

The experimental results show that both models can perform this classification task well, but the CNN model had consistently better performance than the ResNet model according to various metrics. The CNN reached an accuracy of about 94% overall, while the ResNet model got around 91%. While both models showed strong discrimination ability, the CNN outperformed the accuracy, precision, recall, and F1 score metrics.

A key feature of this study is the focused on recall specifically in cancer detection. In medical diagnosis, false negatives — cancers that go undetected — can have dire consequences.) The CNN model achieved a higher cancer class recall than the ResNet model, suggesting that it was better able to identify true positive cancer cases. The ResNet model, on the other hand showed more false negatives and are not reliable in clinical settings.

These results are further validated by confusion matrix analysis. The CNN model had fewer misclassifications and sustained improved sensitivity–specificity balance. As for the ResNet model, though it demonstrated an accuracy in identifying non-cancer cases, it struggled to consistently detect cancer cases. This illustrates the necessity of thoughtful choice of evaluation metrics in medical use cases where accuracy, on its own, does not represent actual performance.

The other major enabler of the work is Grad-CAM for model interpretability. We observe that Grad-CAM visualizations confirm tented model expectations, indicating that the CNN is interpreting information local to tumor regions when making predictions. This quote attention suggests that the model knows on useful and relevant features from data. Conversely, the ResNet model generated

activation maps that were more diffuse and scattered, indicating that this particular architecture plays to broader and less specific features. This greater interpretability gives an additional case for the CNN model.

The other thing the paper does well is offer intuition into when model complexity works. Although ResNet is a more complex architecture with increased depth, its performance was lower than that of the simpler CNN model in this instance. This is largely due to the size and nature of the dataset, which is small and domain-specific. More complex models like ResNet could need more data to truly use their strength but simpler ones can do better if customized properly.

Also, the training and validation results show that there is a more stable convergence and generalization than CNN model. The validation performance of the ResNet model showed oscillations, indicating potential overfitting or susceptibility to changes in data. This suggests not to go for a special architecture just because it seems fancy, but actually choose the model based on your dataset properties.

In conclusion, this study shows that breast cancer detection using ultrasound images can obtain high accuracy, high recall rate and also achieve certain interpretability through the well-designed CNN model. In the context of medical AI, this can help balance performance vs interpretability vs model complexity. This can be seen as a feasible, efficient approach that may provide assistance to medical professionals in the disease detection and diagnosis process.

5.1 Future Work

Although the results of this study are encouraging, there remain many avenues for future improvements and extensions to the proposed method. A fundamental drawback of this work is the dataset size and its diversity. Future studies should integrate larger and more heterogeneous datasets from various clinical sources. With a larger dataset, the models may generalize better and minimize the bias on relevant image features.

The utilization of advanced data augmentation techniques is another area which needs significant future work. Data augmentation techniques such as rotation, flipping, scaling and contrast enhancement increase the variability among the training data and help in extracting more robust features for the model. This is especially useful in the medical imaging context, where collecting large datasets can be difficult.

Another appealing path is the investigation of more advanced deep learning architectures. In fact the recent models like EfficientNet and attention based networks achieved a better performance in

many image classification challenges. The network architectures used can be of that which improves upon feature extractions; and localizability of the tumor regions. Now Hybrid models such as CNNs and attention could improve performance and interpretability.

A possible addition is the combination of multiple modalities. Diagnosis in real-world clinical settings depends not only on imaging data but also patient history, clinical reports and other diagnostic tests. UltraSound images are combined with other clinical information from patient histories to predict outcomes more accurately and completely.

An area with the great needing for future works issue of advancing models interpretability. Grad-CAM is a great first step toward visualizing your model's decision-making, but more advanced techniques may give you better insight into the details. Consider the use of explainees such as SHAP (SHapley Additive exPlanations) or integrated gradients for more details.

Additionally, implementation of the proposed system in real clinical settings is a crucial step toward seeing this tie into practice. A user-friendly interface for clinicians, plus validation in clinical trials, would be needed to ensure reliability and usability. It would also aid in understanding potential pitfalls and further fine-tuning for application in the world outside academic institutions.

Lastly, future research directions may consider real-time implementation and refining the model to achieve accelerated inference. This would allow for integration of the system into clinical workflows to support health professionals in making timely and accurate decisions.

5.2 Reflection

In summary, this project was a great opportunity to learn how to implement deep learning on real-world medical datasets. The project posed several difficulties, including dealing with imbalanced datasets, variability in image quality, and model interpretability. A systematic approach to data preprocessing, model development, and evaluation was essential to tackle these challenges.

There were a million lessons that I gained from this project, but one of them is the importance of exploratory data analysis. Such insights from EDA were essential to gauge the dataset and identify challenges in our way ahead, like class imbalance or varied image dimensions. This knowledge influenced both preprocessing steps and model pipeline design.

Another key learning outcome was a comparison of various model architectures. Interestingly, one of the main hypotheses was that because the ResNet model is deeper and uses transfer learning, it would perform better than a custom CNN. However, the results indicated that with the more basic

CNN model performed better. This emphasized the need to choose models that reflects the qualities of the dataset rather than just picked upon complexity.

I particularly found the implementation of Grad-CAM quite useful, since it offered visualizations on how models reached their predictions. This improved both the interpretability of the models as well as increased confidence in their predictions. Explainability of models is particularly important in medical applications where transparency is essential.

On a technical level, the project helped to hone the skills surrounding deep learning, data pre-processing and model evaluation. It also gave valuable experience with working on real datasets and address issues commonly seen in practice, such as overfitting and model stability. Using evaluation metrics other than accuracy (e.g. recall, ROC-AUC), highlights the need for a holistic approach to the evaluation process.

The project also improved their problem-solving and critical thinking skills. Evaluating how well the model performs and what can be done to improve it needed careful consideration of decisions and further experimentation. It helps me to understand the factors involved in each step and repeating this process is key to forming a good machine learning solution.

This was a very fruitful project where we were able to apply theoretical knowledge and turn it into something usefull. This was a big push for AI in health and laid the groundswell for future experimentation and discoveries.

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7. Appendices

1. Github Repository Link:

[Please click here for GitHub Repository Link For Project Code](#)

2. Dataset Link:

<https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset/data>